

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0026

Roosevelt Rd 138 kV
October 02, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0026. The Interconnection Customer proposes a 133.2 MW solar photovoltaic facility interconnecting at the Roosevelt Rd 138 kV substation in the Virginia area. The request was received on April 19, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 22 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,233,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Innovative Solr 31 Substation-Kapstone 138 KV: 106.9% loading (243 MVA) following the outage of the Louisa Generation Facility-Gimes 138 KV; DFAX = 22.3%.

Pratt'S-Innovative Solar 31 138 KV: 106.0% loading (240 MVA) following the outage of the Louisa Generation Facility-Grimes 138 KV L; DFAX = 22.3%.

For the loss of Louisa Generation Facility Substation-N Anna 345 KV, the Louisa Generation Facility Substation-Carrabelle 345 KV loads to 103.0% of its 1,492 MVA rating. Project DFAX: 10.4%.

Louisa Generation Facility-North Anna 345 KV: 102.1% loading (1,686 MVA) following the outage of the Louisa Generation Facility-Carrabelle 345 KV line; DFAX = 13.6%.

Tombstone-Sharyland 345 KV: 100.3% loading (1,152 MVA) following the outage of the Tombstone-Bixby Hill 500 KV; DFAX = 5.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kapstone-Innovative Solar 31 Substation 138 KV L	Reconductor line	\$2,749,000	100% (\$2,749,000)
Pratt'S-Innovative Solar 31 138 KV	Reconductor line	\$4,066,000	100% (\$4,066,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Louisa Generation Facility-Carrabelle 345 KV	Reconductor line	\$52,929,000	12% (\$6,304,000)
Louisa Generation Facility-North Anna 345 KV	Reconductor line	\$27,716,000	13% (\$3,508,000)
Tombstone-Sharyland 345 KV	Reconductor line	\$15,994,000	10% (\$1,606,000)
	Total allocated to project		\$18,233,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.